

JEE MAIN REVISION MASTER: HALOALKANES AND HALOARENES 2026 3 QUESTIONS 5% WEIGHTAGE

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 19 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Kcet Haloalkanes And Haloarenes - Core Application	19 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Kcet Haloalkanes And Haloarenes - Core Application (Very High Yield — 19 PYQs)

Core Principle: Key problem pattern in Kcet Haloalkanes And Haloarenes.

■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "C-Cl bond in methyl chloride compared to C-Cl bond in chlorobenzene is..." [Ans: D]
- [2026 Shift] Q#2 (MCQ): "In S_N1 reaction, the alkyl halide that on hydrolysis produces racemic mixture is..." [Ans: B]
- [2026 Shift] Q#3 (MCQ): "The compound from which chlorobenzene cannot be prepared easily is..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

C-Cl bond in methyl chloride compared to C-Cl bond in chlorobenzene is

- (A) Longer and stronger
- (B) Shorter and stronger
- (C) Shorter and weaker
- (D) Longer and weaker **[Correct]**

Correct Answer: D

Solution Analysis: The C-Cl bond in methyl chloride is compared with the C-Cl bond in chlorobenzene. In methyl chloride, CH_3Cl , carbon is sp^3 hybridised and the C-Cl bond is a normal single bond. In chlorobenzene, $\text{C}_6\text{H}_5\text{Cl}$, the chlorine atom is attached to an sp^2 hybridised carbon of the benzene ring. Also, due to resonance, the lone pair of chlorine overlaps with the benzene ring. This gives the C-Cl bond some partial double bond character. Because of this partial double bond character, the